MODULE 4.3

Configure ACLs

Access Control Lists | IPv4 Packet Filtering

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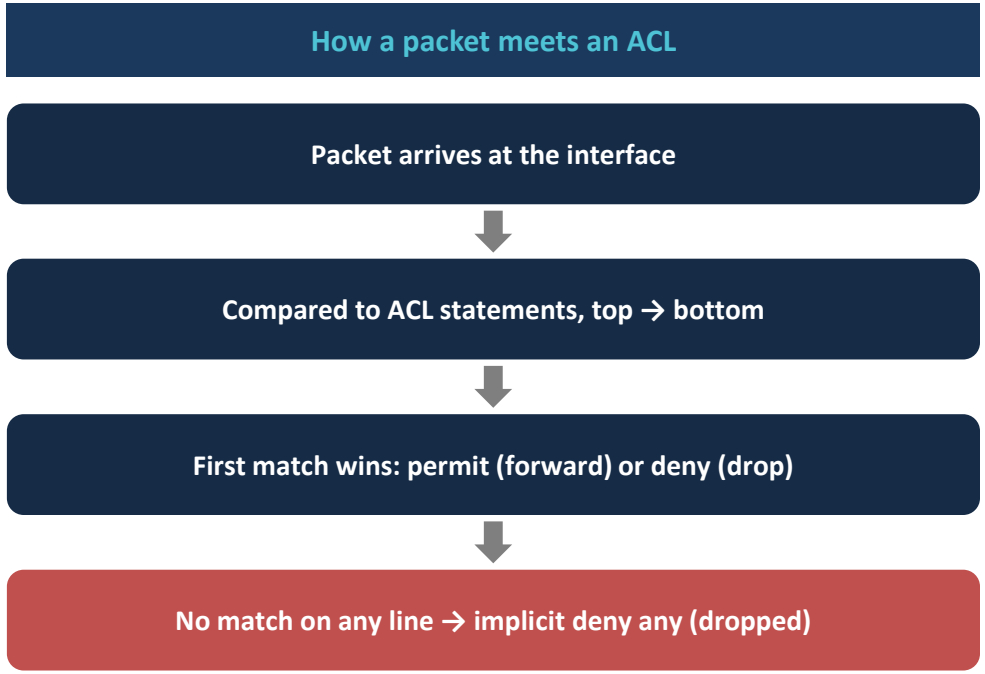
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Create an ACL

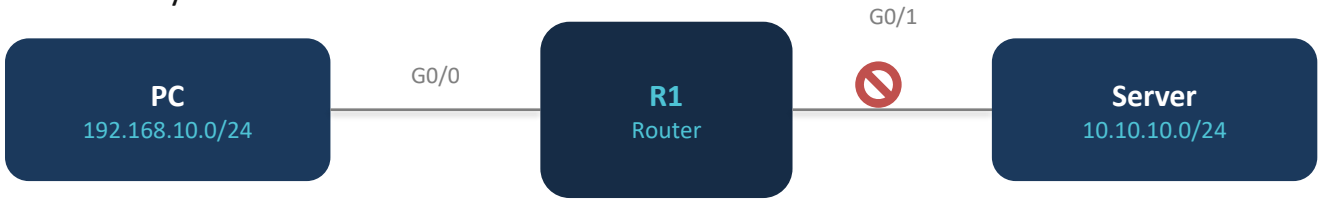
- **Two-step process:** first create the ACL by defining a numbered or named list of permit/deny statements; second, activate it on an interface, inbound or outbound, with the ip access-group command (IPv4).
- **Numbered vs. named:** a numbered ACL (access-list 10 ...) is quick to type but anonymous; a named ACL (ip access-list standard BLOCK_GUEST) documents its own purpose and lets you edit individual lines later without rebuilding the whole list.
- **Standard vs. extended:** standard ACLs filter on source address only, so apply them close to the destination; extended ACLs filter on source, destination, protocol, and port, so apply them close to the source.
- **Processing order:** every statement is checked top to bottom, and the router stops at the first match — specific lines belong above general ones, and an invisible "deny all" sits at the end of every ACL.



Abbreviation: ACL = Access Control List.

Numbered Standard IPv4 ACL Syntax

- **Syntax:** access-list access-list-number {permit | deny} source [source-wildcard] — standard ACLs only ever examine the source IP address, nothing else.
- **Number range:** standard IPv4 ACLs use 1–99, or the expanded range 1300–1999; the router reads the number itself to know you mean "standard," not "extended."
- **Wildcard mask:** the inverse of a subnet mask — a 0 bit means "must match," a 1 bit means "don't care." So 0.0.0.255 against 192.168.10.0 matches any host in 192.168.10.0/24.
- **Placement rule:** because a standard ACL can't see the destination, apply it as close to the destination as possible, or you risk blocking that source from reaching networks you never meant to restrict.



Standard ACL 10 blocks 192.168.10.0/24, applied outbound on G0/1 — close to the destination.

```
          Applied on R1, closest to destination:
R1(config)# access-list 10 deny 192.168.10.0 0.0.0.255
R1(config)# access-list 10 permit any
R1(config)# interface g0/1
R1(config-if)# ip access-group 10 out
```

```
                              Verify:
R1# show access-lists 10
```

Abbreviations: ACL = Access Control List. IPv4 = Internet Protocol version 4.

Named Standard IPv4 ACL Syntax

- **Step 1 — create and name it:** ip access-list standard NAME drops you into a named-ACL sub-configuration mode, prompt becomes (config-std-nacl)#, ready for permit/deny lines.
- **Step 2 — add statements:** {permit | deny} {source [source-wildcard] | any} — identical matching logic to a numbered standard ACL, just without a number out front.
- **Naming convention:** names are case-sensitive, can't contain spaces, and by convention are written in ALL_CAPS with underscores, e.g. BLOCK_GUEST_WIFI, so the name documents its own job.
- **Editing later:** every line has an implied sequence number, so you can remove one line with "no permit ..." or insert a new line at a chosen position — something a numbered ACL can't do without a full rebuild.

Create and apply:

```
R1(config)# ip access-list standard BLOCK_GUEST
R1(config-std-nacl)# deny 192.168.10.0 0.0.0.255
R1(config-std-nacl)# permit any
R1(config-std-nacl)# exit
R1(config)# interface g0/1
R1(config-if)# ip access-group BLOCK_GUEST out
```

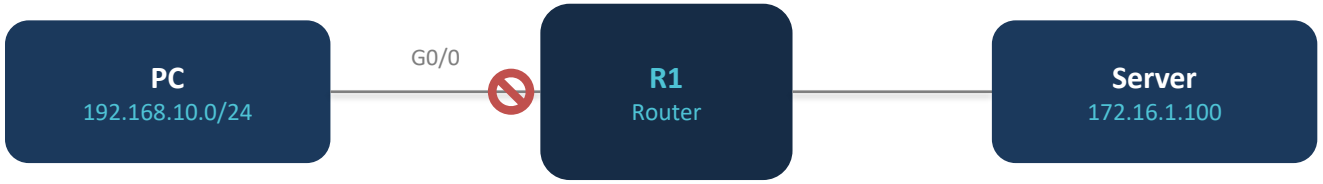
Verify:

```
R1# show access-lists BLOCK_GUEST
```

Abbreviation: ACL = Access Control List.

Numbered Extended IPv4 ACL Syntax

- **Syntax:** access-list access-list-number {permit|deny} protocol source [wildcard] [operator port] destination [wildcard] [operator port] [established] — three extra fields versus standard: protocol, destination, and optional port operators.
- **Number range:** extended IPv4 ACLs use 100–199, or the expanded range 2000–2699, so the router instantly knows this list can filter on more than source address.
- **Protocol field:** usually ip (all IP traffic), tcp, udp, or icmp — naming a specific protocol here is what unlocks port-matching keywords later in the same line.
- **Placement rule:** because extended ACLs can already see the destination, apply them as close to the source as possible — it stops unwanted traffic at the first hop instead of letting it cross the network first.



Extended ACL 110 blocks web traffic from this PC, applied inbound on G0/0 — close to the source.

```
                Applied on R1, closest to source:
R1(config)# access-list 110 deny tcp
    192.168.10.0 0.0.0.255 host 172.16.1.100 eq 80
R1(config)# access-list 110 permit ip any any
R1(config)# interface g0/0
R1(config-if)# ip access-group 110 in
```

Abbreviation: IP = Internet Protocol.

Protocols and Port Numbers

- **Operator keywords:** eq (equal to), neq (not equal to), lt (less than), gt (greater than), and range (an inclusive span of two ports) let an extended ACL match a specific port instead of a whole protocol.
- **Name or number:** Cisco lets you type a service name instead of its number for common ports — eq www works exactly like eq 80.
- **Which address it follows matters:** the port operator goes right after whichever address it filters — a source port after the source address, a destination port after the destination address.

Protocol	Port(s)	Transport
FTP	20, 21	TCP
SSH	22	TCP
Telnet	23	TCP
SMTP	25	TCP
DNS	53	TCP / UDP
DHCP	67, 68	UDP
HTTP	80	TCP
POP3	110	TCP
HTTPS	443	TCP

Abbreviations: FTP=File Transfer Protocol, SSH=Secure Shell, SMTP=Simple Mail Transfer Protocol, DNS=Domain Name System, DHCP=Dynamic Host Configuration Protocol, HTTP(S)=Hyper Text Transfer Protocol (Secure), POP3=Post Office Protocol v3.

Port Number Configuration Examples

- **eq — match exactly:** access-list 103 permit tcp any host 192.168.10.5 eq 80 allows traffic only to port 80 (HTTP) on that one server, nothing else.

- **range — match a span:** access-list 104 ... range 20 21 allows both FTP ports in a single line instead of two statements — only works because 20 and 21 are contiguous.

- **lt / gt — match below or above:** access-list 105 ... lt 1024 allows any well-known port (0–1023) in one line, handy for permitting "all the standard services" at once.

Examples on R1:

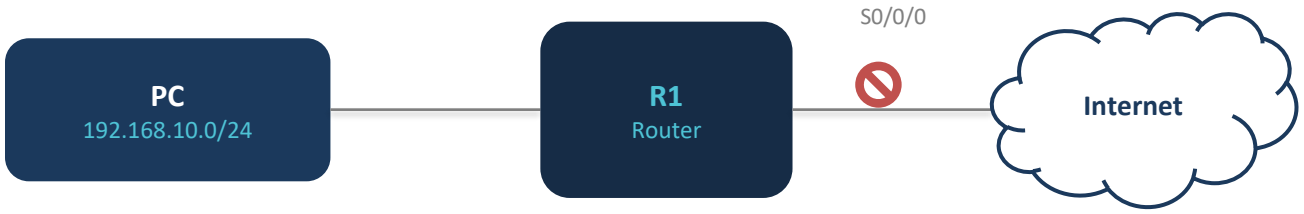
```
! Allow only HTTP to the server (eq)
R1(config)# access-list 103 permit tcp any
    host 192.168.10.5 eq 80

! Allow both FTP ports in one line (range)
R1(config)# access-list 104 permit tcp any
    host 192.168.10.5 range 20 21

! Allow every well-known port below 1024 (lt)
R1(config)# access-list 105 permit tcp any
    host 192.168.10.5 lt 1024
```

TCP Established Extended ACL

- **The problem:** denying all inbound internet traffic also blocks return traffic for sessions internal hosts started — like the webpage replies to a browser request.
- **The established keyword:** matches a TCP segment only if its ACK or RST flag is set — segments belonging to a session someone inside already opened, never a brand-new inbound attempt.
- **What it doesn't do:** established is just a header check, not a stateful firewall — a lightweight filter, not a substitute for proper stateful inspection.
- **This topology:** hosts on 192.168.10.0/24 browse outward freely; nothing from the internet may start a new connection inward.



New inbound connections are blocked; replies to PC-initiated sessions are permitted.

```
                Inbound on S0/0/0 (ACL 120):
R1(config)# access-list 120 permit tcp any
    192.168.10.0 0.0.0.255 established
R1(config)# access-list 120 deny tcp any any
```

```
                                Applied:
R1(config-if)# ip access-group 120 in
```

Abbreviations: TCP = Transmission Control Protocol. ACK = Acknowledgment flag. RST = Reset flag. SYN = Synchronize flag.

Named Extended IPv4 ACL Syntax

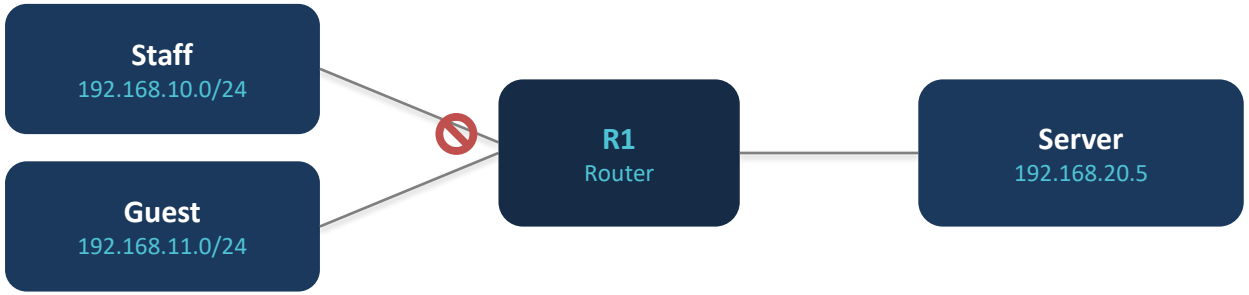
- **Step 1 — create and name it:** ip access-list extended NAME enters extended named-ACL sub-configuration mode, prompt becomes (config-ext-nacl)#.
- **Step 2 — add statements:** {permit|deny} protocol source [wildcard] [operator port] destination [wildcard] [operator port] [established] — the same fields as the numbered version, just no number first.
- **Why bother naming it:** a name like NO_FTP_OUT tells the next admin exactly what the list does, where a bare "access-list 110" forces them to read every line to work out its purpose.
- **Same editing benefit:** as with named standard ACLs, individual lines can be removed or inserted by sequence number without rebuilding the whole ACL from scratch.

```
                    Create and apply:
R1(config)# ip access-list extended NO_FTP_OUT
R1(config-ext-nacl)# deny tcp 192.168.10.0
    0.0.0.255 any eq 21
R1(config-ext-nacl)# deny tcp 192.168.10.0
    0.0.0.255 any eq 20
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)# exit
R1(config)# interface g0/0
R1(config-if)# ip access-group NO_FTP_OUT in
```

Abbreviation: FTP = File Transfer Protocol (control port 21, data port 20).

Named Extended IPv4 ACL Example

- **Scenario:** Staff (192.168.10.0/24) need full access to the server at 192.168.20.5; Guest (192.168.11.0/24) should reach only its web service on port 80, nothing else.
- **Named ACL GUEST_WEB_ONLY** is built and applied inbound on R1's interface facing the Guest subnet — closest to the source, consistent with the extended-ACL placement rule from 4.3.4.
- **Verification:** a ping from a Guest host to any other service on the server should fail, while a Guest browser session to the server on port 80 should succeed — proving the ACL is scoped, not a blanket block.



Guest reaches only TCP port 80 on the server; all other Guest traffic to it is denied.

Create and apply:

```
R1(config)# ip access-list extended GUEST_WEB_ONLY
R1(config-ext-nacl)# permit tcp 192.168.11.0
    0.0.0.255 host 192.168.20.5 eq 80
R1(config-ext-nacl)# deny ip 192.168.11.0
    0.0.0.255 host 192.168.20.5
R1(config-ext-nacl)# permit ip any any
R1(config)# interface g0/2
R1(config-if)# ip access-group GUEST_WEB_ONLY in
```

Verify:

```
R1# show access-lists GUEST_WEB_ONLY
```

Summary

- ACLs enforce one golden rule: permit only what's necessary, and let the implicit deny handle everything else.
- **4.3.1–4.3.4 —** every ACL is created first, then applied to an interface; standard ACLs (1–99 / 1300–1999) filter on source only, extended ACLs (100–199 / 2000–2699) filter on source, destination, protocol, and port.
- **4.3.5–4.3.6 —** operator keywords (eq, neq, lt, gt, range) let extended ACLs match specific ports, not just whole protocols.
- **4.3.7 —** the established keyword lets return traffic back in without opening the network to new inbound connections.
- **4.3.8–4.3.9 —** named ACLs document themselves and can be edited line by line, making them the preferred style for anything beyond a quick test.

References

Cisco Networking Academy — Network Defense, Module 4.3: Configure ACLs

https://www.netacad.com

Cisco — Configuring IP Access Lists

https://www.cisco.com/c/en/us/support/docs/ip/access-lists/26448-ACLsamples.html

Cisco — Guide to Harden Cisco IOS Devices

https://www.cisco.com/c/en/us/support/docs/ip/access-lists/13608-21.html